



Institutional Knowledge Embedding for Generative Artificial Intelligence-Assisted Decision Support in University Traffic Safety Management

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Abstract: University traffic safety management acceleratingly relies on digital systems, yet frontline decisions continue to pose challenges when rules are dispersed across independent departments. Situations are composite as responsibilities often overlap. This study developed and analytically assessed an institutional knowledge embedding (IKE) framework for generative artificial intelligence (AI)-assisted decision support in university traffic safety management. A design-oriented conceptual research method was adopted, combining an integrative synthesis of higher education governance, institutional theory, organizational knowledge, socio-technical systems, accountability, and retrieval-augmented generation (RAG) research with structured scenario analysis. The analysis produced five linked governance mechanisms: normative encoding, contextual qualification, responsibility allocation, evidentiary inscription, and reflexive updating. These mechanisms were translated into a structured institutional knowledge unit and a RAG-enabled multi-agent architecture in which AI supported retrieval, comparison, workflow drafting, and evidence organization, while authorized human actors retained final judgment. Structured comparison across four university traffic-safety scenarios demonstrated that the framework became most valuable not when it automated decisions, but when it made explicit rule validity, contextual conditions, responsible roles, evidentiary requirements, and escalation points. The study further derived an institutional non-closure rule: when valid grounds, applicable rules, or authorized responsibility could not be established, the system should abstain from determinate recommendations and transfer the case to human review. The framework contributes to further research on educational technology, university governance, and campus safety management by linking generative AI adoption to the improvement of institutional accountability, procedural traceability, and controlled organizational learning.

Keywords: Generative artificial intelligence; University governance; Campus safety; Institutional knowledge embedding; Decision support; Retrieval-augmented generation

1. Introduction

Generative artificial intelligence (AI) is progressing exponentially from conversational assistance toward organizational decision support. In higher education, this transition is visible not only in teaching and learning, but also in administrative work, policy analysis, student services, risk management, and institutional governance. Recent research has emphasized both the transformative potential of generative AI and the requirement for governance arrangements that help maintain the visibility of educational values, human responsibility, and institutional rules during the adoption of relevant technology (Bond et al., 2024; Chan, 2023; Crompton & Burke, 2023). Within university management, large language models could retrieve regulations, summarize complex documents, draft procedures, and support cross-departmental coordination. However, these capabilities generate a fundamental management question to be resolved in the current paper: How could AI-assisted recommendations remain connected to the institutional rules, role boundaries, evidentiary requirements, and review processes that render university decisions legitimate and accountable?

University traffic safety is a potential arena which offers a compact yet analytically rich setting for examining this question. Campus gates, roads, parking areas, event venues, and fire lanes are spaces under the management of tertiary institutes in which public service, safety, mobility, emergency response, and organizational responsibility intersect. Four recurring structural problems complicate frontline handling in this setting: rules are dispersed across vehicle-management policies, visitor procedures, parking rules, fire-safety requirements, event plans, and temporary notices. The apparently similar events could belong to different scenarios; responsibility is distributed among gate officers, patrol staff, duty managers, school representatives, event organizers, and property personnel. Evidence is scattered across telephone calls, messaging applications, access-control logs, paper forms, photographs, and work orders. These complicated conditions elevate campus traffic management into something more than a technical access-control problem, hence the emergence of rule-intensive university governance problem.

Within higher education management, campus traffic safety serves not merely as the security or transport function. It is an administrative service that coordinates student and staff mobility, visitor access, emergency readiness, event organization, property operations, and institutional responsibility. Decisions related to the management of gates and roads on campus could reveal how university rules are interpreted, how various departments cooperate, and how service and safety obligations are balanced. Handling traffic safety, particularly in an educational setting, is conducive to examining how digital governance principles are translated into everyday university administration.

Research on digital transformation in higher education has increasingly emphasized that adoption of technology should be accompanied by institutional adaptation. Chinese scholarship on university governance, for example, argued that digital technologies could improve governance capacity only when value, technology, and institutional arrangements were aligned (Li et al., 2024; Wu & Wu, 2024). Similarly, research on holistic governance in the generative-AI era highlighted the need to address organizational fragmentation and cross-departmental coordination rather than treating AI as an isolated tool (Hou & Guan, 2025). At the international level, socio-technical research warned that AI performance and accountability emerged from interactions among models, users, organizations, interfaces, procedures, and feedback loops instead of from model capability alone (Baxter & Sommerville, 2011; Theodorou & Dignum, 2020).

A related technical development is retrieval-augmented generation (RAG), which enables language models to retrieve external information at inference time (Lewis et al., 2020). Dense passage retrieval improves semantic matching across differently worded documents (Karpukhin et al., 2020), while self-reflective retrieval approaches could facilitate models to assess whether additional evidence is indispensable (Asai et al., 2024). These techniques are beneficial for upholding university regulations because relevant provisions may be distributed across multiple documents. Nevertheless, semantic relevance is not equivalent to institutional validity. A retrieved paragraph may be obsolete, temporary, overridden by a higher-level rule, inapplicable to the current actor, or contingent on an exception. Similarly, a multi-agent workflow may divide technical tasks among specialized modules without clarifying who is institutionally authorized to verify facts, approve exceptions, execute actions, or bear the responsibility.

This gap is especially important for education management because university digitalization is not simply an efficiency project. It also redistributes authority, standardizes judgment, produces records, and shapes what becomes organizational knowledge. Recent work in Education Science and Management has highlighted that large language models could improve the efficiency and consistency of university governance-text analysis when they operate within an explicitly analytical workflow (Qin et al., 2025). In this connection, the present study extended this direction from policy-text analysis to frontline decision support and argued what institutional structure should accompany AI-generated advice when organizational action, safety risk, and accountability were involved.

To bridge the gap of study, the following three research questions (RQs) were addressed:

RQ1: How could fragmented university rules, responsibilities, procedures, scenarios, and evidence requirements be structured as machine-readable but institutionally meaningful knowledge?

RQ2: How could RAG and multi-agent coordination support the processes of university safety decision without transferring final authority from human actors to AI systems?

RQ3: What sorts of governance requirements and boundary conditions would emerge when the framework was examined across routine, ambiguous, high-risk, and rule-conflict scenarios?

Reflecting on the above, this research made three contributions. First, it introduced institutional knowledge embedding (IKE) as a design concept for higher education management and distinguished it from document digitization, conventional knowledge management, and ordinary RAG. Second, it developed a five-mechanism model encompassing normative encoding, contextual qualification, responsibility allocation, evidentiary inscription, and reflexive updating. The model was then translated into a structured knowledge unit and socio-technical architecture. Third, it derived an institutional non-closure rule and a set of future evaluation indicators through analytical scenario testing. The objective of the present study is not to claim that the proposed system has already been deployed or empirically validated. Rather, it provided a research framework that could be implemented and tested in later university pilots.

2. Literature Review and Theoretical Basis

This section establishes the conceptual foundations of the study by connecting four bodies of literature that were directly relevant to rule-intensive university management. It first reviewed digital transformation and university governance, then examined generative AI in higher education management, organizational knowledge and accountability, and RAG-based multi-agent systems. Taken together, these streams identified the governance gap addressed by IKE: technical access to information does not by itself ensure institutional validity, authorized action, procedural traceability, or accountability.

2.1 Digital Transformation and University Governance

Digital transformation in higher education has expanded from teaching platforms and learning analytics to institutional management, policy implementation, and governance. This expansion changes the object of educational technology research. When technology enters admission, personnel, safety, compliance, or resource allocation, its effects could not be assessed only through usability or efficiency. Researchers should also examine how organizational rules, authority, responsibility, and public-service obligations were represented in digital systems. Chinese scholars of digital university governance emphasized that effective transformation required coordination among technological capacity, institutional arrangements, and educational values (Li et al., 2024; Wu & Wu, 2024). Holistic governance research further contended that generative AI could reduce informational fragmentation though it may generate new coordination and responsibility problems without a clear demarcation of organizational boundaries (Christensen & Lægreid, 2007; Hou & Guan, 2025).

This perspective is consistent with broader institutional research. Scott (2014) described institutions as combinations of regulative, normative, and cultural-cognitive elements that stabilized organizational behavior. Formal rules therefore acquired practical meaning through responsibilities, routines, accepted exceptions, and shared expectations. Similarly, Granovetter (1985)'s concept of embeddedness rejected the idea that action could be understood as isolated calculation. When applied to university management, a rule such as "unregistered vehicles may not enter" could not be interpreted unambiguously without comprehending the rule's authority, effective period, recognized exceptions, responsible verifier, and escalation route.

2.2 Generative Artificial Intelligence in Higher Education Management

Previous research on generative AI in education has largely focused on teaching, learning, assessment, academic integrity, and student adoption. At the governance level, however, universities are commencing to explore large language models for policy analysis, administrative communication, knowledge search, and decision support. Chan (2023) proposed an AI policy education framework for university teaching and learning, thus emphasizing the need for institutional policies alongside technology adoption. Qin et al. (2025) showed how a large-language-model workflow could support policy-instrument analysis of university governance texts. More recent institutional studies added that university generative AI guidance extended well beyond classroom use to administration, security, ethics, accountability, human oversight, and infrastructure (An et al., 2025). Evidence gathered from these studies indicated that AI could increase the accessibility and analytical use of institutional documents, but they did not eliminate the necessity for human validation and governance controls.

A technology-institution-value perspective further suggested that technical capability should be interpreted in combination with institutional arrangements and educational values (Li et al., 2024). Generative AI could not only accelerate information retrieval, text processing, and workflow coordination, but it could also produce hallucinations, privacy risks, algorithmic bias, overreliance, and responsibility gaps (An et al., 2025; Bond et al., 2024). In university traffic safety management, these risks are particularly consequential because certain actions would involve restricting access, emergency coordination, incident escalation, or the handling of personal data. An appropriate question about educational management is therefore not enquiring whether AI could generate an answer, but whether the organization could justify and review the action that derives from that answer.

2.3 Organizational Knowledge and Institutional Accountability

Organizational knowledge theory provides a second foundation. Nonaka (1994) explained knowledge creation as a dynamic process through which tacit and explicit knowledge interacted. University safety work contains both forms. Formal policies, emergency plans, and job descriptions are explicit, while frontline judgment about exceptions, sequencing, and coordination is often tacit. However, a simple conversion of tacit experience into searchable text is insufficient in a rule-intensive domain because experience does not automatically possess normative authority. IKE therefore adds a governance layer to knowledge conversion: operational experience should be reviewed before it becomes reusable institutional knowledge.

Accountability theory incorporates the organizational requirement that responsibility should remain assignable.

Accountable AI could not be achieved by merely treating the algorithm as the responsible actor; organizations should identify who verifies, authorizes, acts, and answers for consequences (Busuioc, 2021; Novelli et al., 2024). Meaningful human control further requires that reviewers be informed, authorized, and able to change the outcome (Santoni de Sio & Van den Hoven, 2018; Verdiesen et al., 2021). For IKE, this means that the retrieved content should be accompanied by the role, evidence, and review information required for institutional accountability.

2.4 Retrieval-Augmented Generation, Multi-Agent Systems, and the Governance Gap

RAG reduces dependence on a language model’s parametric memory by linking generation to external knowledge (Lewis et al., 2020). For university governance, this is useful because rules change over time and may be distributed across units. Yet, retrieval systems generally optimize semantic similarity or evidence relevance, whereas institutional decision support requires source hierarchy, version status, jurisdiction, exceptions, procedural sequence, and responsibility. The same problem appears in agentic architectures. A scenario-recognition agent, retrieval agent, responsibility-matching agent, and drafting agent may improve modularity, but the distribution of computational tasks does not establish a legitimate distribution of organizational authority.

Socio-technical systems theory therefore provides the integrating perspective. Baxter & Sommerville (2011) treated system performance as the result of interactions between technical and social components. Shneiderman (2020) argued that high automation could coexist with high human control when systems are designed around human-centered reliability. Steen et al. (2025) cautioned against images of AI as a free-standing autonomous entity and emphasized the dependence of AI on people, institutions, and feedback loops. In the present study, RAG and multi-agent coordination were therefore treated as implementation technologies within a broader institutional arrangement rather than as governance mechanisms on their own.

To make the governance contributions explicit, Table 1 compares simple document storage, conventional RAG, and the proposed IKE framework. The comparison revealed that IKE supplemented institutional properties that were not normally guaranteed by document access or semantic retrieval alone.

Table 1. Comparison of document storage, conventional retrieval-augmented generation (RAG), and institutional knowledge embedding (IKE)

Dimension	Document Storage	Conventional RAG	IKE
Rule validity	Stores files/metadata; validity is checked manually	Retrieves indexed content; current authority is not guaranteed	Checks source authority, effective dates, supersession, and version status
Context/Exceptions	Interpreted manually	May retrieve relevant passages but may miss applicability	Represents applicability conditions, exceptions, conflicts, and missing context
Responsibility	Usually outside the repository	May assign technical tasks	Specifies verifier, approver, executor, reviewer, and accountable role
Evidence	Documents remain available but decision chains may be fragmented	May cite sources without recording human confirmation or action	Records grounds, relevant facts, human confirmation, deviations, and review path
Human approval	External to storage	Optional step of the workflow	Required before AI output becomes an organizational decision
Feedback	Manual replacement or archiving	Updated content could be re-indexed	Only reviewed, authorized, and versioned feedback may revise authoritative knowledge

3. Methodology

This section explains how the proposed framework was developed and analytically investigated. Rather than evaluating an already deployed system, the study adopted a design-oriented conceptual approach that combined problem decomposition, integrative theoretical synthesis, conceptual construction, and structured scenario assessment. The following subsections specify the research design, analysis procedures, analytical criteria, and the ethical and validity limits of the study.

3.1 Research Design

This study adopted a design-oriented conceptual research approach. It did not just report an implemented campus system, participant survey, experiment, or operational dataset. Instead, it developed a theoretically grounded governance framework and examined its internal logic through structured analytical scenarios. This design was appropriate to the research objective because the study asked what institutional information and

decision controls should be represented before empirical system evaluation could become meaningful. The research output was therefore a set of design constructs, a knowledge schema, a socio-technical architecture, and analytically derived governance requirements that could later be operationalized in field studies.

The university traffic-safety setting was used as a bounded higher-education management domain because it combined routine administrative service, interdepartmental coordination, emergency preparedness, public-facing decision making, and institutional accountability within a single operational environment.

3.2 Procedures for Analysis

The analysis proceeded in four stages. First, problem decomposition was conducted using university traffic-safety management as the focal governance context. Four recurring management deficits were retained as the problem frame: dispersed rules, composite scenarios, overlapping responsibilities, and discontinuous evidence. These deficits were treated as design requirements rather than as measured frequencies. Second, an integrative literature synthesis connected the problem frame to higher education governance, institutional theory, organizational knowledge creation, accountability and human oversight, socio-technical systems, and RAG research. The synthesis was theory driven rather than a systematic review; its purpose was to identify concepts capable of explaining why document retrieval alone was insufficient for rule-intensive educational management.

Third, conceptual construction translated the theoretical synthesis into five governance mechanisms and a minimal institutional knowledge unit. Each mechanism was required to answer a distinct management question: Is the source valid? Does it apply here? Who is authorized? What must be recorded? How may experience change future knowledge? Fourth, structured scenario assessment examined the framework across four contrasting university traffic-safety vignettes: an unregistered visitor vehicle, congestion during a major event, a vehicle obstructing a fire lane, and a conflict between temporary and standing rules or a case of institutional silence. The scenarios were deliberately varied in rule clarity, risk, and responsibility complexity.

The four scenarios were selected purposively rather than sampled empirically. Together, they formed a small analytical matrix that varied three dimensions: rule clarity, operational risk, and responsibility complexity. The unregistered-visitor case represents relatively clear rules with common exceptions and low-to-medium risk; major-event congestion represents plan-dependent rules with medium risk and high coordination needs; fire-lane obstruction combines comparatively clear rules with high operational risk; and rule conflict or institutional silence represents low rule clarity and high interpretive complexity. The scenarios are hypothetical but management-realistic vignettes constructed from recurrent university traffic-management task types identified in the problem decomposition. They do not reproduce confidential incidents, operational logs, or a specific university case database.

3.3 Analytical Criteria

Each scenario was examined using five criteria derived from the framework: normative traceability, contextual fit, responsibility clarity, evidentiary completeness, and escalation appropriateness. Normative traceability asks whether the recommendation can be connected to a competent and current institutional source. Contextual fit asks whether relevant conditions and exceptions have been identified. Responsibility clarity asks whether verification, approval, execution, and review roles are explicit. Evidentiary completeness asks whether the decision episode can be reconstructed and contested. Escalation appropriateness asks whether the system stops or transfers control when institutional grounds, authority, or risk conditions do not support determinate generation. These criteria provide a transparent analytical basis for comparing the scenarios without claiming empirical effect sizes.

3.4 Validity and Limitations

This conceptual study used no human participants, personal records, or operational university datasets. The scenario analysis was illustrative and did not represent observed incidents. Therefore, the validity of the results was analytical rather than statistical. The framework was evaluated for conceptual coherence, coverage of the four identified governance deficits, and consistency across contrasting scenarios. Empirical validation is required before drawing conclusions regarding decision quality, efficiency, user acceptance, or safety outcomes.

4. Results

This section reports the principal analytical outputs generated through the research procedure. It first defines IKE and specifies the structure of an institutional knowledge unit, then derives the five governance mechanisms and translates them into a socio-technical architecture for AI-assisted decision support. The section subsequently compares four university traffic-safety scenarios and derives implementation and evaluation indicators for future empirical testing.

4.1 Definition and Structure of Institutional Knowledge Embedding

The first result was the definition of IKE as a socio-technical process through which institutionally valid rules, applicability conditions, role boundaries, procedural requirements, evidentiary obligations, and revision controls were transformed into machine-readable and organizationally actionable knowledge while final interpretive authority and responsibility remained with authorized human actors. The concept differs from ordinary digitization because it does not treat the document as the final knowledge object. It differs from general knowledge management because it preserves normative force and authorization. It differs from ordinary RAG because it requires validity, exceptions, roles, evidence, and controlled revision to derive from the retrieved content.

The unit of analysis is an institutionally grounded recommendation episode rather than an isolated model output. Such an episode begins with a situated request, connects the request to valid institutional grounds, identifies applicable conditions and responsible roles, records evidence and human confirmation, and determines whether any subsequent experience is eligible for review and knowledge-based revision. Table 2 presents the minimal structure of an institutional knowledge unit.

Table 2. Minimal schema for an institutional knowledge unit

Field	Content	Purpose of Governance
Normative source	Document title, clause, issuing authority, organizational status	Establishes provenance and authority
Validity and version	Effective date, expiry date, superseded or temporary status	Prevents obsolete or provisional rules from being treated as current
Application domain	Visitor access, event management, contractor access, emergency passage, fire lanes	Limits inappropriate transfer across situations
Triggering conditions	Actor or vehicle type, reservation status, time, place, operational status, risk level	Connects rules to observable circumstances
Exceptions and conflicts	Emergency exceptions, superior rules, competing provisions, unresolved ambiguity	Prevents over-rigid or decontextualized application
Authorized roles	Information provider, verifier, recommender, approver, executor, reviewer	Preserves the institutional division of labor
Procedural actions	Register, wait, contact, divert, authorize, issue work order, review, archive	Converts norms into sequenced organizational tasks
Evidence requirements	Logs, photographs, calls, confirmations, timestamps, reasons, appeal information	Supports answerability and contestation
Feedback status	Operational note, isolated exception, reviewed precedent, approved revision	Prevents unreviewed experience from becoming policy

4.2 Five Governance Mechanisms

The second result was a five-mechanism model. Normative encoding converts institutional materials into structured grounds for action and preserves source hierarchy, validity, and version status. Contextual qualification determines whether a valid rule applies to the present situation and makes missing facts, exceptions, or rule conflicts explicit. Responsibility allocation identifies who may verify, recommend, approve, execute, and review. Evidentiary inscription records the rules, facts, human confirmation, reasons for deviation, and available review pathway. Reflexive updating separates operational memory from authoritative knowledge so that raw logs and exceptional decisions do not automatically become future policy.

The mechanisms are analytically distinct but sequentially connected. A source that is valid but contextually inapplicable should not support action; a contextually appropriate rule without an authorized decision maker should not become an organizational decision; and a decision that could not be reconstructed through evidence is weakly accountable. Likewise, feedback that bypasses review could convert local workarounds into silent institutional drift. Figure 1 exhibits this governance cycle.

4.3 Socio-Technical Architecture for Generative Artificial Intelligence-Assisted Decision Support

The third result was a layered architecture that positioned AI mediation between institutional knowledge infrastructure and human decision authority. The institutional environment contains rules, mandates, role definitions, procedural safeguards, and review provisions. These are represented in a versioned knowledge infrastructure. The AI mediation layer performs scenario interpretation, hybrid retrieval, conflict detection, responsibility mapping, and draft generation. The human decision layer provides contextual judgment, authorization, modification, rejection, and escalation. The evidence and learning layer records the decision episodes and routes that could review feedback toward knowledge revision.

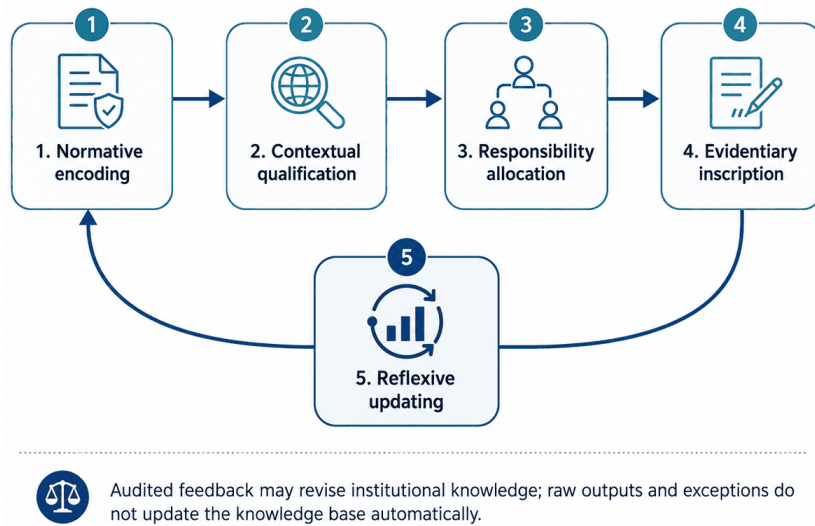


Figure 1. Institutional knowledge embedding (IKE) as a governance cycle

Within the AI layer, specialized agents may be assigned to scenario recognition, rule retrieval, responsibility matching, recommendation drafting, and evidence packaging. A coordinating agent may compare outputs and return inconsistent results for further retrieval. However, computational delegation does not transfer institutional authority. The architecture therefore allocates cognitive labor asymmetrically: machines search, compare, summarize, monitor versions, and structure records; humans interpret ambiguity, weigh competing objectives, authorize exceptions, and remain answerable for final organizational actions. Privacy and security, contestability, auditability, and oversight operate as cross-cutting controls rather than as optional post-processing steps.

Operationally, the scenario-recognition agent extracts the vehicle or actor type, location, time, reservation or event status, and risk cues from the request. The rule-retrieval agent searches only current and competent institutional sources and returns the relevant clause, version status, applicability conditions, and known exceptions. The responsibility-matching agent maps the case to the university roles permitted to verify facts, approve exceptions, execute actions, and review disputes. The evidence agent specifies the minimum records required to justify and reconstruct the decision, while the drafting agent combines the validated outputs into a conditional recommendation that clearly marks unresolved issues. A coordinating agent checks consistency across these outputs. If agents return conflicting information, the coordinator requests a second retrieval or flags the conflict for human review; if no current institutional rule could be found, the system enters institutional non-closure and could not convert the absence of a rule into a determinate recommendation. As illustrated in Figure 2, these components form a layered socio-technical architecture that connects institutional knowledge infrastructure, AI-mediated processing, authorized human decision making, and auditable organizational learning.

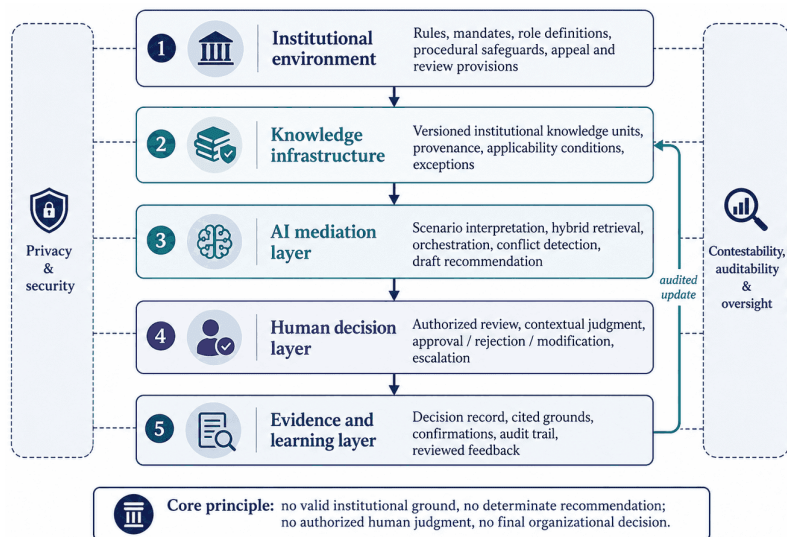


Figure 2. Socio-technical architecture for institutionally grounded generative artificial intelligence (AI)

4.4 Scenario-Based Analytical Results

As summarized in Table 3, the four scenarios reveal how the contribution of AI changes with rule clarity, operational risk, and responsibility complexity. The four scenarios reveal how the contribution of AI changes with rule clarity and risk. In the unregistered-visitor scenario, the main value is contextual qualification and responsibility routing. In event congestion, the system supports coordination by retrieving the approved event plan and clarifying which actions are already authorized and which require approval. In a fire-lane obstruction, high risk increases the need for rapid human control even when the rule is clear. In rule conflict or institutional silence, the correct system behavior is not to force a single answer but to expose the unresolved institutional condition.

This comparison produced a key governance finding: technical confidence should not be treated as institutional confidence. A highly similar retrieved document may still be normatively weak, while a low-frequency but authoritative emergency rule may be decisive. The analysis therefore derived an institutional non-closure rule: when valid grounds, applicability conditions, or authorized responsibility could not be established, the system should not issue a determinate organizational recommendation. It may request clarification, present competing provisions, summarize uncertainty, and escalate the case to an authorized human forum.

Table 3. Analytical comparison of four university traffic-safety scenarios

Scenario	Rule Clarity/Risk	Primary Artificial Intelligence (AI) Contribution	Required Human Role*	Escalation Trigger
Unregistered visitor	Usually high with exceptions/Low-medium	Scenario questions; current rule retrieval; host-confirmation workflow; record template	Verifier: gate officer. Approval/exception: duty security supervisor or host unit under local rules. Execution: gate/security staff. Dispute review: designated security manager.	Forced entry; missing confirmation; disputed exception
Major-event congestion	Plan-dependent/Medium	Retrieve event plan; compare activation triggers; present authorized options; coordinate evidence	Verifier: event organizer or traffic-duty officer. Approval: Security office duty manager. Execution: gate, patrol, property, or event staff. Review: event/security lead.	Expired or missing plan; emergency access threatened
Fire-lane obstruction	High/High	Rapid rule and role retrieval; evidence checklist; high-priority incident workflow	Verifier: patrol or gate staff. Approval: duty security supervisor. Execution: designated security/property staff. Review: campus safety manager.	Immediate safety threat; contested status; unavailable authority
Rule conflict or silence	Low/Variable	Expose competing sources, hierarchy, missing policy, and provisional alternatives	Verifier: responsible unit and security staff. Provisional approval: authorized security/administrative manager. Execution: designated unit. Review: policy owner or competent review body.	No competent rule; unresolved hierarchy; novel technology

Note: *Role titles differ across universities; the roles shown below are functional examples and should be mapped to each institution's formal delegation and job descriptions.

Figure 3 summarizes the resulting decision workflow. The workflow began with incident or request parsing, proceeded through rule and role retrieval, and introduced an explicit decision point regarding the sufficiency of institutional grounds. If grounds were sufficient, the system might draft a recommendation and evidence package for authorized human review. However, if they were insufficient, the workflow would proceed to institutional non-closure. Approved actions were recorded and reviewed feedback might be returned to the knowledge-governance process rather than directly updating the knowledge base.

In order to better explain the research approach of this article, an example of “unregistered visitor vehicles” was used for illustration below:

Step 1—Input. A gate officer enters a request such as: “A visitor vehicle has arrived without a reservation and has stated that it is attending a university meeting”.

Step 2—Scenario recognition. The system identifies an unregistered visitor, the current gate and time, the stated purpose, and the absence of a reservation, while marking emergency status as not indicated.

Step 3—Rule retrieval. The rule-retrieval agent searches the current visitor-access policy and, for illustration, retrieves the active requirement that an unregistered visitor must receive host or designated-unit confirmation before entry; it also checks whether an event-specific exception is in force.

Step 4—Verification and responsibility. The gate officer verifies the plate and stated purpose, while the responsibility-matching agent identifies the host unit or duty security supervisor as the role permitted to confirm or approve an exception.

Step 5—Draft recommendation and evidence. If the rule and confirmation are consistent, the system drafts a conditional workflow: move the vehicle to the waiting area, contact the host, record the confirmation, and then allow or refuse access after human authorization. The evidence record contains only the information required by university policy.

Step 6—Escalation. If the host cannot be confirmed, the applicable rule is missing or conflicting, or if the vehicle refuses to wait or attempts forced entry, the system stops routine generation of recommendation and transfers the case to the duty security supervisor.

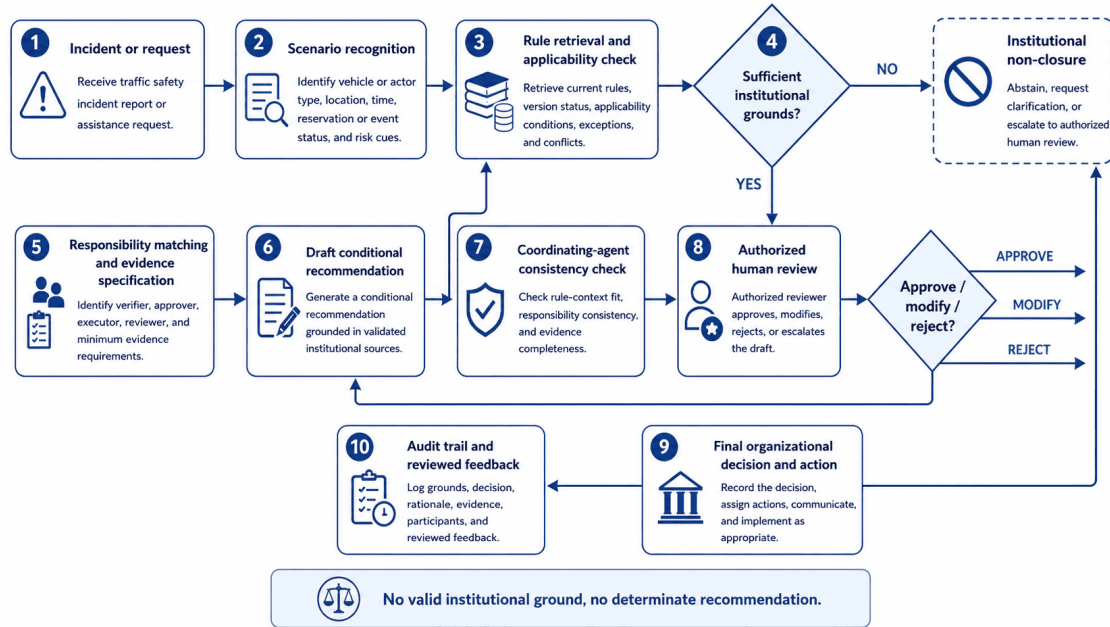


Figure 3. Institutionally grounded decision workflow for university traffic safety assistance

4.5 Derived Implementation and Evaluation Indicators

The framework also produced a set of indicators for future empirical implementation. To move evaluation beyond response speed toward institutional reliability, the indicators were categorized into four categories: grounding quality, process quality, human-control quality, and learning quality. These indicators were not measured in the present study; they were proposed as dependent variables for later pilots. The proposed indicators and their corresponding interpretations are summarized in Table 4.

Table 4. Proposed indicators for future empirical evaluation

Dimension	Candidate Indicator	Interpretation
Grounding quality	Institutional-source hit rate; rule-version accuracy; exception-detection rate	Measures whether recommendations are connected to valid and current grounds
Process quality	Procedure-standardization rate; responsibility-matching accuracy; evidence-chain completeness	Measures whether advice is operationally structured and traceable
Human-control quality	Human-confirmation completeness; escalation-trigger accuracy; override availability	Measures whether human review is substantive rather than ceremonial
Learning quality	Reviewed-feedback adoption rate; rule-drift incidents; repeated unresolved escalations	Measures whether organizational learning is controlled and whether policy gaps become visible

5. Discussion

This section interprets the analytical results in relation to higher education management and responsible AI governance. It discusses how the framework extends existing research on university digital governance, clarifies the roles of human authority and institutional non-closure, translates the findings into practical implications for university administrators, and identifies the principal risks and boundary conditions that would constrain implementation.

5.1 Contributions to Higher Education Management

The results repositioned generative AI in university management from a document-answering technology to a component of institutional infrastructure. This mattered because educational management research often evaluated digital systems through adoption, efficiency, satisfaction, or performance, whereas rule-intensive administrative decisions also required legitimacy, role clarity, and procedural traceability. The IKE framework connected these concerns by showing how institutional knowledge should move through a complete decision episode. In this light, the study extended the value-technology-institution logic discovered in university digital-governance research (Li et al., 2024; Wu & Wu, 2024; Zhang, 2024), the holistic-governance emphasis on cross-unit coordination (Hou & Guan, 2025), and recent international work demanding institution-wide generative AI-governance that includes human oversight, accountability, privacy, and continuing policy review (An et al., 2025).

The study also extended AI-supported educational governance from institutional text analysis to situated administrative action. While Qin et al. (2025) shed light on how large language models could support structured analysis of university governance texts, IKE would focus on what should accompany those texts to inform an operational decision: validated applicability, responsible roles, evidence, and review.

5.2 Human Authority and Institutional Non-closure

A central implication is that “human in the loop” is insufficient when the human role is only ceremonial. In the proposed framework, the reviewer should have timely information, relevant competence, institutional authority, and a practical ability to reject or revise the recommendation (Santoni de Sio & Van den Hoven, 2018; Subías-Beltrán et al., 2026). Human approval is therefore treated as an institutional control rather than a final interface click.

The institutional non-closure rule complements that control. When rules conflict, authority is missing and a novel or high-risk case exceeds the available institutional basis. The system should abstain, expose the unresolved condition, and escalate it. Repeated non-closure could then be treated as a management signal that policy, delegation, or emergency procedures require revision.

5.3 Practical Implications for University Administrators

For university administrators, implementation should begin with knowledge governance rather than model procurement. Institutions should implement inventory policies, identify issuing authorities, record effective dates and supersession relationships, define application domains, specify exceptions, and map verification and approval roles before connecting documents to a RAG system. Temporary event plans and local workarounds should be tagged explicitly so that semantic similarity does not cause them to be retrieved as general rules.

Second, universities should design escalation by risk and reversibility. Routine wayfinding or parking-information requests may support broad automated assistance, while cases involving emergency access, fire lanes, forced entry, penalties, or sensitive personal information require stronger verification and human control. A universal numeric confidence threshold is therefore inappropriate. Model confidence is a technical property whereas institutional confidence depends on source authority, contextual completeness, risk, and the availability of a competent decision maker.

Third, evidence governance should balance traceability with privacy. For a routine visitor-access case, the minimum necessary record may include the vehicle plate, entry time and gate, visitor or host confirmation status, the decision reason, and the identity or role of the human reviewer. Telephone numbers, identity-document details, photographs, call records, or other personal information should be collected only when required by an applicable rule or by the risk of the incident. Access should be role based: gate staff should view the information only needed for current verification; duty supervisors and designated reviewers may access the full case record when required for approval, audit, or dispute resolution; and system administrators should not receive routine content access solely because they maintain the technical platform. Identifiers should be deleted or anonymized once the operational, appeal, audit, or legally required retention purpose has expired, while de-identified event metadata may be retained for approved learning and policy review. Any longer retention should follow applicable university policy and law rather than the convenience of the AI system.

Fourth, knowledge updating should be separated from routine operational logging. Incident summaries, model outputs, and exceptional approval should enter an isolated review space. Only validated, de-identified where necessary, authorized, and versioned material should modify the long-term knowledge base. This separation protects organizational learning from feedback contamination and rule drift while preserving the possibility of double-loop learning in which repeated incidents trigger formal policy change rather than silent adaptation.

5.4 Risks and Boundary Conditions

The framework has several boundary conditions. First, institutional completeness matters. A knowledge base

could not repair absent, contradictory, or deliberately ambiguous rules; it could only expose the gap. Second, over-formalization may privilege what is easy to encode and marginalize tacit expertise, service values, accommodation needs, or legitimate discretion. Third, institutional validity is not identical to normative legitimacy. An accurately embedded rule may still be unfair or disproportionate. University governance should therefore preserve appeal, contestation, exception review, and ethical evaluation rather than treating institutional grounding as sufficient proof of justice. This concern is consistent with recent higher-education research showing that generative AI governance should address ethics, equity, and the framing effects of institutional policy rather than focusing only on compliance or efficiency (Bond et al., 2024; Luo, 2024).

Fourth, automation bias and deskilling remain important, alongside broader risks identified in recent research on generative AI in higher education (An et al., 2025; Bond et al., 2024). Proficient and well-cited recommendations may appear more authoritative than they are, leading staff to defer the system or lose competence in rule interpretation. Interfaces should therefore show uncertainty, alternative grounds, unresolved questions, and the identity of the responsible human actor (Miller, 2019). Fifth, feedback processes may be captured by powerful organizational groups. Frontline concerns, students' or visitors' objections, and minority perspectives may disappear if only managerial interpretations enter the knowledge loop. Reflexive updating should therefore preserve dissent and enable review by more than one organizational role.

6. Conclusions

This study examined how generative AI could support university traffic safety management when frontline decisions depended on dispersed institutional rules, composite scenarios, overlapping responsibilities, and fragmented evidence. Using design-oriented conceptual research and structured scenario analysis, it developed an IKE framework that linked normative encoding, contextual qualification, responsibility allocation, evidentiary inscription, and reflexive updating.

The analytical results showed that the value of generative AI in this setting lied less in autonomous decision making than in institutional orientation. AI could help locate valid rules, identify missing contextual information, structure workflows, match responsible roles, assemble evidence, and reveal when escalation is necessary. An RAG-enabled multi-agent architecture could support these functions, but it remains only one technical layer within a larger socio-technical arrangement in which human actors retain interpretive authority and accountability.

The most significant governance implication in this study is the rule of institutional non-closure: when valid institutional grounds, applicability conditions, or authorized responsibility could not be established, the system should not produce a determinate organizational recommendation. This principle provided a practical way to distinguish responsible assistance from unauthorized automation. It also turned uncertainty into a management signal by making policy gaps, conflicts, and delegation problems visible.

The study was limited by its conceptual design and scenario-based analytical assessment. It did not demonstrate improvements in decision accuracy, processing time, user satisfaction, or safety outcomes. Future research should implement the proposed knowledge schema in a limited university pilot, compare conventional document retrieval with IKE-informed RAG, and evaluate institutional-source accuracy, process standardization, evidence completeness, escalation accuracy, human overreliance as well as user perceptions. Comparative studies across campus safety, student services, laboratory management, and other rule-intensive university functions would further test the generalizability of the framework.

A practical first pilot could focus on the low-risk unregistered-visitor scenario at one campus gate, excluding emergencies, forced-entry incidents, penalties, and other high-risk cases. Gate officers would use the interface for rule retrieval and evidence prompts, while one duty supervisor could provide authorized review for exceptions or unresolved cases. During a defined trial period, the pilot could compare the existing document/manual workflow with IKE-informed assistance using three initial indicators: rule-version accuracy, responsibility-matching accuracy, and evidence-chain completeness. The pilot should also record all escalations and overrides qualitatively so that errors could be reviewed before the framework was extended to higher-risk scenarios.

Author Contributions

Conceptualization, Q.H. and Z.H.W.; methodology, Q.H. and Z.H.W.; formal analysis, Q.H. and Z.H.W.; writing—original draft preparation, Z.H.W.; writing—review and editing, Q.H. and Z.H.W. All authors have read and agreed to the published version of the manuscript.

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Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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